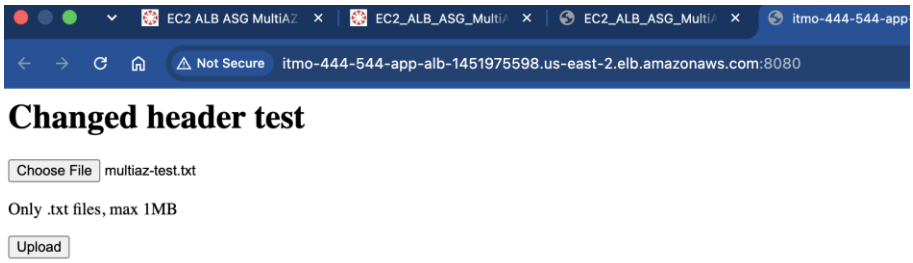
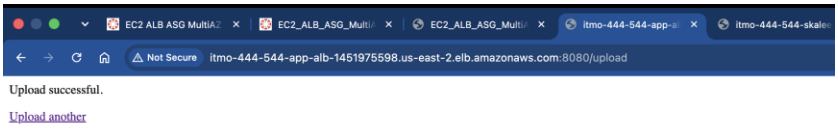
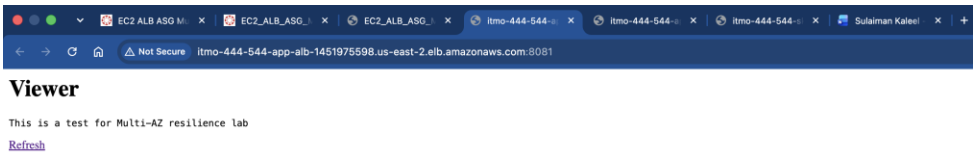
Sulaiman Kaleel - EC2_ALB_ASG_MultiAZ_Resilience

1.
```
== Building user-data from local lab03/ files ==
== Building launch template data ==
== Creating/updating launch templates ==
== Looking up the default VPC and subnets in both AZs ==
== ALB security group ==
== Tightening the app security group to only accept traffic from the ALB ==
== Target groups ==
== Application Load Balancer ==
Waiting for the ALB to become active...
== Listeners ==
== Auto Scaling Groups ==
Done. ALB DNS name: itmo-444-544-app-alb-1451975598.us-east-2.elb.amazonaws.com
Uploader: http://itmo-444-544-app-alb-1451975598.us-east-2.elb.amazonaws.com:8080/
Viewer: http://itmo-444-544-app-alb-1451975598.us-east-2.elb.amazonaws.com:8081/
Give the target groups 1-2 minutes to report healthy.
Check with:
aws elbv2 describe-target-health --target-group-arn arn:aws:elasticloadbalancing:us-east-2:589354718744:targetgroup/itmo-444-544-uploader-tg/014a8241082a27da --region us-east-2
[root@c75863f757b0:/AWS_CLI_EC2# /var/folders/fj/xwnqtb452zb6tlmrps2b84xm0000gn/T/TemporaryItems/NSIRD_screencaptureui_dy0W^C
[root@c75863f757b0:/AWS_CLI_EC2# ^C
root@c75863f757b0:/AWS_CLI_EC2#
```



2.





3.

```
root@c75863f757b0:/AWS_CLI_EC2# aws elbv2 describe-target-health \
--target-group-arn "$UPLOADER_TG_ARN" \
--region us-east-2 --no-cli-pager
{
    "TargetHealthDescriptions": [
        {
            "Target": {
                "Id": "i-068d2c4c37b9d3287",
                "Port": 3000
            },
            "HealthCheckPort": "3000",
            "TargetHealth": {
                "State": "healthy"
            },
            "AdministrativeOverride": {
                "State": "no_override",
                "Reason": "AdministrativeOverride.NoOverride",
                "Description": "No override is currently active on target"
            }
        }
    ]
}
root@c75863f757b0:/AWS_CLI_EC2# aws elbv2 describe-target-health \
--target-group-arn "$VIEWER_TG_ARN" \
--region us-east-2 --no-cli-pager
{
    "TargetHealthDescriptions": [
        {
            "Target": {
                "Id": "i-0f763cb28567bfed0",
                "Port": 3000
            },
            "HealthCheckPort": "3000",
            "TargetHealth": {
                "State": "healthy"
            },
            "AdministrativeOverride": {
                "State": "no_override",
                "Reason": "AdministrativeOverride.NoOverride",
                "Description": "No override is currently active on target"
            }
        }
    ]
}
```

4.

```
root@c75863f757b0:/AWS_CLI_EC2# python3 testing/chaos_test.py --tier uploader
Target: uploader tier at http://itmo-444-544-app-alb-1451975598.us-east-2.elb.amazonaws.com:8080/
Confirming the endpoint is healthy before we start...
Terminating i-068d2c4c37b9d3287 via the Auto Scaling Group...
Waiting for the endpoint to go down...
Waiting for the endpoint to come back up...
Poll log written to chaos_test_uploader_4778.csv
Recovery time: 78.0s (SLO: 180s)
PASS: recovery time met the SLO.
root@c75863f757b0:/AWS_CLI_EC2#
```

```
root@c75863f757b0:/AWS_CLI_EC2# python3 testing/chaos_test.py --tier viewer --slo-seconds 120
Target: viewer tier at http://itmo-444-544-app-alb-1451975598.us-east-2.elb.amazonaws.com:8081/
Confirming the endpoint is healthy before we start...
Terminating i-0f763cb28567bfed0 via the Auto Scaling Group...
Waiting for the endpoint to go down...
Waiting for the endpoint to come back up...
Poll log written to chaos_test_viewer_4978.csv
Recovery time: 65.5s (SLO: 120s)
PASS: recovery time met the SLO.
root@c75863f757b0:/AWS_CLI_EC2#
```

5.

```
root@c75863f757b0:/AWS_CLI_EC2# python3 testing/chaos_test.py --tier uploader --slo-seconds 60
Target: uploader tier at http://itmo-444-544-app-alb-1451975598.us-east-2.elb.amazonaws.com:8080/
Confirming the endpoint is healthy before we start...
Terminating i-07d4896e77912e99b via the Auto Scaling Group...
Waiting for the endpoint to go down...
Waiting for the endpoint to come back up...
Poll log written to chaos_test_uploader_5674.csv
Recovery time: 71.7s (SLO: 60s)
FAIL: recovery time exceeded the SLO.
root@c75863f757b0:/AWS_CLI_EC2#
```

6. Using my recovery time information, it was reported as 78.0 seconds (uploader) with the 180 second SLO and 65.5 seconds for the viewer with the 120 second SLO. Once the timing was changed to 60 seconds, the uploader failed with 71.7 seconds. Looking at the results, the npm's installation as well as the app initial startup is what contributed mostly. In terms of health checks, those also added additional time as well.

7. Comparing the last lab and this one, looking at create_appstack.sh, in that lab there were run instances used which then would launch 2 separate instances (EC2), however in this lab instead that was replaced with Auto scaling groups. These use launch templates for instance instead. I would say because when there is a failed instance it is able to replace it automatically.

8.

```
root@c75863f757b0:/AWS_CLI_EC2# ./delete_app_stack.sh
== Deleting Auto Scaling Groups ==
Waiting for both Auto Scaling Groups to be gone...
itmo-444-544-uploader-asg deleted.
itmo-444-544-viewer-asg deleted.
== Deleting the load balancer ==
== Deleting target groups ==
== Restoring the app security group to its previous-lab state ==
== Deleting the ALB security group ==
{
    "Return": true,
    "GroupId": "sg-0a0264c7842366686"
}
== Deleting launch templates ==
{
    "LaunchTemplate": {
        "LaunchTemplateId": "lt-0267bdc5182fb604c",
        "LaunchTemplateName": "itmo-444-544-uploader-lt",
        "CreateTime": "2026-11-12T00:37:07+00:00",
        "CreatedBy": "arn:aws:iam::589354718744:user/sulaiman-kaleel",
        "DefaultVersionNumber": 1,
        "LatestVersionNumber": 1,
        "Operator": {
            "Managed": false
        }
    }
}
{
    "LaunchTemplate": {
        "LaunchTemplateId": "lt-0f790843ba3ea9d76",
        "LaunchTemplateName": "itmo-444-544-viewer-lt",
        "CreateTime": "2026-11-12T00:37:09+00:00",
        "CreatedBy": "arn:aws:iam::589354718744:user/sulaiman-kaleel",
        "DefaultVersionNumber": 1,
        "LatestVersionNumber": 1,
        "Operator": {
            "Managed": false
        }
    }
}
== Removing IAM instance profiles and roles ==
== Emptying and deleting S3 bucket ==
delete: s3://itmo-444-544-skaleel-fileshare/shared.txt
Cleanup complete.
root@c75863f757b0:/AWS_CLI_EC2#
```